

Itay Yona

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Professional Experience

- 2022-Present **AI21 Labs (NLP), Algorithm Developer**
Developing State-of-the-Art Deep Learning NLP capabilities for our Large Language Models
Enhancing the reliability, quality, and general applicability of LLMs
- 2021-2022 **Israeli Ministry of Defense, Research Scientist & AI Consultant**
Designed, conducted and analyzed information from controlled laboratory-based investigations, experiments and trials on Deep Neural Networks
- 2020-2022 **Weizmann AI Center, Machine Learning Researcher**
Developed a novel Causal-Discovery ML approach to multicollinear data by combining multiple Explainable ML methods: GAMs, Bayesian statistics, Shapely values
- 2018-2019 **Pagaya (Fintech), Senior Data Scientist**
Initiated from scratch the real estate research, yielding 30% of the company revenues
- 2016–2018 **8200 Unit, Israeli Defense Forces (NSA/GCHQ equivalent)**
Senior Researcher - Cyber-Security & Electrical Engineering
Solved the flagship project of the unit by bridging two seemingly unrelated disciplines
- The project was awarded the Israeli Defense Prize of 2021
- 2016 **AHAM Course Instructor**
Commanded and tutored highly talented mathematicians and cyber-security specialists
- 2013–2016 **Researcher - Cyber-Security & Math**
Directed state-of-the-art interdisciplinary mathematical and cryptographic systems research

Education

- 2020-2022 **Weizmann Institute of Science, M.Sc. Computer Science, Math & Neuroscience**
Thesis: Analysis of High-Dimensional Neuronal Tuning Curves in Bat Hippocampus during Complex Behaviors. Advisor: Prof. Nachum Ulanovsky
Towards publication (in progress)
- Presented on: SFN (2021), ISFN (2021), Cosyne (2022)
 - Awarded full 2-years research fellowship
- 2016-2019 **The Open University, B.Sc. Computer Science**
During the military service
- Graduated Magna Cum Laude
 - University Presidential Award in 2018
- 2013 **AHAM Intelligence Course, Research Track**
Completed prestigious exclusive 6 month trainings for (the nationally top 15) exceptional mathematicians and tech prodigies

Volunteering

2021-2023 The Problem-Solving Course, Co-Founder

Lecturing gifted students from the Israeli periphery and tech companies
Designed course material on the principles of problem-solving

2022-Present Effective Altruism Israel, Coordinator of AI explainability forum

Initiated a reading and hackathon group for mechanistic interpretability of Deep Neural Networks. It has over 50 scholars and 10 permanent participants from varied disciplines: Cybersecurity, Neuroscience, and Artificial Intelligence

Awards

Won 1st place in 5 major international cyber-security CTF tournaments (team Pasten):

- 32C3 (2015)
- 33C3 (2016)
- Google CTF (2016)
- 34C3 (2017)
- Google CTF (2017)

Technical Summary

Programming Skills: Python, C\C++, x86 Asm

Professional Interests: Deep Learning, Machine Learning, Algorithms, Statistics, Neuroscience, Natural Language Processing, Reinforcement Learning, Math, Software and Hardware Reverse Engineering

Frameworks: PyTorch, TensorFlow, Kubernetes